

Online Appendix

Substitute or Complement? Generative AI Use and Student Achievement: Evidence from PISA
2025

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This appendix accompanies the paper. Appendix A reports the exact estimates behind the paper’s figures, and Appendix B shows that they hold under every definition of inattentive responding. Appendix C shows them under two further blocks of controls. Appendices D to G document the questionnaire items, the construction of the index, the response-style artifact, and the sample.

A Main estimates

Tables A.1 and A.2 report the exact estimates and standard errors behind Figures 2 and 4 of the paper, together with the coefficients on the controls and the fit statistics that the figures omit. Table A.3 re-estimates the second stage of Section 7 behind Figure 5 under alternative weights, standard errors and samples, and Table A.4 estimates the same relationships within countries at the school level.

Table A.1: AI use and achievement: full regression results behind Figure 2

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Mathematics</i>					
AI-use index	-4.08 (0.32)	-1.26 (0.33)	-3.67 (0.31)	-3.78 (0.31)	-5.77 (0.24)
Socioeconomic status (ESCS)			33.91 (0.38)	33.52 (0.37)	16.38 (0.23)
Straight-lining indicator				-13.05 (0.87)	-7.43 (0.78)
Within-student SD of the items				29.78 (4.89)	17.85 (4.41)
Within-student SD, squared				-23.70 (3.62)	-14.82 (3.31)
R ²	0.002	0.099	0.202	0.209	0.432
F statistic	163.7	14.5	4,183.3	1,802.4	1,111.2
Mean of outcome	476.8	476.8	476.8	476.8	476.8
<i>Panel B. Reading</i>					
AI-use index	-6.11 (0.33)	-3.30 (0.33)	-5.53 (0.31)	-5.72 (0.31)	-7.57 (0.25)
Socioeconomic status (ESCS)			31.34 (0.39)	30.79 (0.39)	13.35 (0.29)
Straight-lining indicator				-17.81 (1.00)	-11.54 (0.88)
Within-student SD of the items				37.47 (5.17)	24.82 (4.64)
Within-student SD, squared				-25.89 (3.97)	-16.75 (3.59)
R ²	0.004	0.065	0.146	0.158	0.379
F statistic	353.2	100.4	3,322.2	1,705.8	763.9
Mean of outcome	477.3	477.3	477.3	477.3	477.3
<i>Panel C. Science</i>					
AI-use index	-6.33 (0.30)	-3.49 (0.32)	-5.94 (0.30)	-6.10 (0.30)	-7.94 (0.22)
Socioeconomic status (ESCS)			34.56 (0.38)	34.09 (0.38)	16.66 (0.26)
Straight-lining indicator				-14.70 (0.87)	-8.96 (0.74)
Within-student SD of the items				35.90 (4.38)	23.22 (3.75)
Within-student SD, squared				-26.36 (3.30)	-16.99 (2.85)
R ²	0.004	0.076	0.178	0.187	0.405
F statistic	435.7	117.8	4,224.5	1,971.7	1,190.3
Mean of outcome	497.8	497.8	497.8	497.8	497.8
Country fixed effects	No	Yes	Yes	Yes	Yes
School fixed effects	No	No	No	No	Yes
Students	249,077	249,077	249,077	249,077	249,077
Schools	10,403	10,403	10,403	10,403	10,403
Countries	35	35	35	35	35

Notes: Weighted least squares of each domain's score on the regressors shown, on the same students in every column, senate-weighted with within-sample renormalization. Standard errors in parentheses combine 80 Fay replicate weights with Rubin's imputation variance across ten plausible values. Columns (2) to (4) add country fixed effects, socioeconomic status and the response-style terms cumulatively; column (5) replaces country with school fixed effects, which subsume them. The F statistic is a design-based Wald test that the coefficients shown are jointly zero. R² is averaged over plausible values and in column (5) includes the school fixed effects. Constants and fixed effects are omitted. The AI-use coefficients are those plotted in Figure 2 of the paper.

Table A.2: The association by kind of AI use: exact estimates behind Figure 4

	Country fixed effects			School fixed effects		
	Math	Reading	Science	Math	Reading	Science
<i>Panel A. The four chatbot items entered together</i>						
Summarize a text	-5.93 (0.35)	-8.38 (0.39)	-8.14 (0.37)	-5.99 (0.31)	-8.47 (0.33)	-8.23 (0.33)
Preliminary research	3.21 (0.38)	2.87 (0.41)	3.54 (0.41)	2.85 (0.30)	2.57 (0.34)	3.22 (0.34)
Draft writing	-7.58 (0.38)	-10.50 (0.36)	-9.80 (0.36)	-5.51 (0.33)	-7.88 (0.32)	-7.43 (0.32)
Help me learn	6.32 (0.40)	10.03 (0.42)	7.87 (0.40)	2.18 (0.32)	5.45 (0.36)	3.57 (0.31)
<i>Panel B. Principal components, orthogonal by construction</i>						
PC1: level (how much)	-3.67 (0.31)	-5.55 (0.31)	-5.95 (0.30)	-5.68 (0.24)	-7.42 (0.25)	-7.82 (0.22)
PC2: purpose (learn vs. produce)	7.15 (0.26)	10.53 (0.27)	9.18 (0.25)	4.44 (0.23)	7.46 (0.23)	6.33 (0.21)

Notes: Exact estimates plotted in Figure 4 of the paper. PISA points per standard deviation, 249,077 students in 35 countries. Items are standardized with senate weights and entered together, so each coefficient holds the other three fixed. All columns include socioeconomic status and the response-style controls; the left block adds country fixed effects and the right block school fixed effects. Standard errors combine the 80 replicate weights with the imputation variance.

Table A.3: System-level moderators of the AI gradient under alternative estimators and specifications

	Purpose slope on teacher support	Level slope on AI use in the country	Level slope on school-based use
<i>Panel A. Alternative estimators</i>			
Precision weights, robust errors (paper)	5.33 (1.47)	15.63 (2.70)	-6.02 (2.38)
Precision weights, classical errors	5.33 (2.07)	15.63 (2.47)	-6.02 (2.79)
Random-effects weights	5.14 (1.61)	16.46 (2.91)	-5.78 (2.44)
Unweighted	5.10 (1.66)	16.64 (2.95)	-5.79 (2.48)
Leave one country out, range	4.58 to 5.79	15.40 to 19.43	-7.56 to -4.96
Spearman rank correlation	0.37 ($p = 0.03$)	0.79 ($p < 0.01$)	-0.43 ($p = 0.05$)
<i>Panel B. Alternative specifications, precision weights</i>			
Adding how common AI use is	5.23 (1.47)		
Adding mean achievement as well	5.91 (1.72)	11.97 (2.71)	-6.48 (2.83)
First-step slopes among AI users only	5.39 (1.51)	11.90 (2.33)	-5.38 (2.27)
Countries	34	35	22

Notes: Mathematics. Each column regresses the within-school first-step slopes of Section 7 of the paper, one per country, on the system characteristic in its head; the last column also conditions on the country mean of the AI-use index, as in the paper. Panel A varies the estimator of that regression. Panel B keeps the paper's precision weights and adds regressors cumulatively, or re-estimates the first-step slopes among students who use AI; a blank cell means the regressor is already in the specification. Precision weights are the inverse of the squared first-step standard error; random-effects weights add the between-country variance of the slopes (DerSimonian and Laird, 1986). Standard errors in parentheses are heteroskedasticity-robust except in the second row of Panel A. The range row drops one country at a time under random-effects weights.

Table A.4: School-level moderators of the AI gradient within countries

	Math	Reading	Science	Students
Purpose component $\times$ teacher support, school mean	-0.86 (0.67)	-0.73 (0.82)	-1.11 (0.75)	249,077
Purpose component $\times$ teacher support, own report	0.28 (0.21)	0.05 (0.22)	0.25 (0.20)	249,077
Level component $\times$ AI use, school mean	3.35 (0.92)	5.38 (0.96)	4.02 (0.95)	249,077
Level component $\times$ school-based use, school mean	0.59 (0.70)	1.28 (0.85)	1.20 (0.84)	162,307

Notes: Coefficient on the interaction between the component and the moderator, in PISA points per standard deviation of the component per unit of the moderator, from student-level regressions with school fixed effects, socioeconomic status and the paper's response-style controls, in which both components and both interactions enter together. The school-mean moderator is the leave-one-out mean among the student's schoolmates, demeaned within country, so its unit matches the country means of Table A.3; the own-report row uses the student's own teacher-support score. Standard errors combine 80 replicate weights with the imputation variance across ten plausible values. The last row uses the 24 countries with the technology questionnaire.

B Robustness to definitions of inattentive responding

The response-style controls of Section 3 of the paper are built from the four chatbot items: a straight-lining indicator and the within-student dispersion of the answers. A control that is a function of the regressor invites the question of whether the estimates depend on it. This appendix answers it by measuring inattention from time on screen and from other parts of the questionnaire, independently of the AI items, and re-estimating the level and purpose coefficients under each.

The questionnaire provides four kinds of evidence on attention, and the check rests on all four, since PISA 2025 fields no dedicated attention-check item. The first is time. The assessment's timing file records the seconds each student spent on every screen, including the one holding the four chatbot items. The median is 19 seconds; a student who leaves the screen within five seconds is too quick to have read four stems, and eight seconds is generous. We flag students under five and under eight seconds, the fastest tenth within each country, and, for the questionnaire as a whole, the fastest twentieth and tenth within each country (Huang et al., 2012).

The second is the pattern of answers elsewhere in the questionnaire. Five batteries mix positively and negatively worded statements, on belonging at school, perseverance, curiosity, beliefs about science and views of school, so the same answer to every statement contradicts itself; on the two five-point batteries the neutral midpoint is exempt. We count such contradictions and flag students with one or more and with two or more. We also compute the share of 27 attitude and frequency batteries a student straight-lines, flagging half or more and three quarters or more, and the shares of answers at the ends and at the midpoint of the scale, flagging the top five percent within country (Krosnick, 1991; Meade and Craig, 2012).

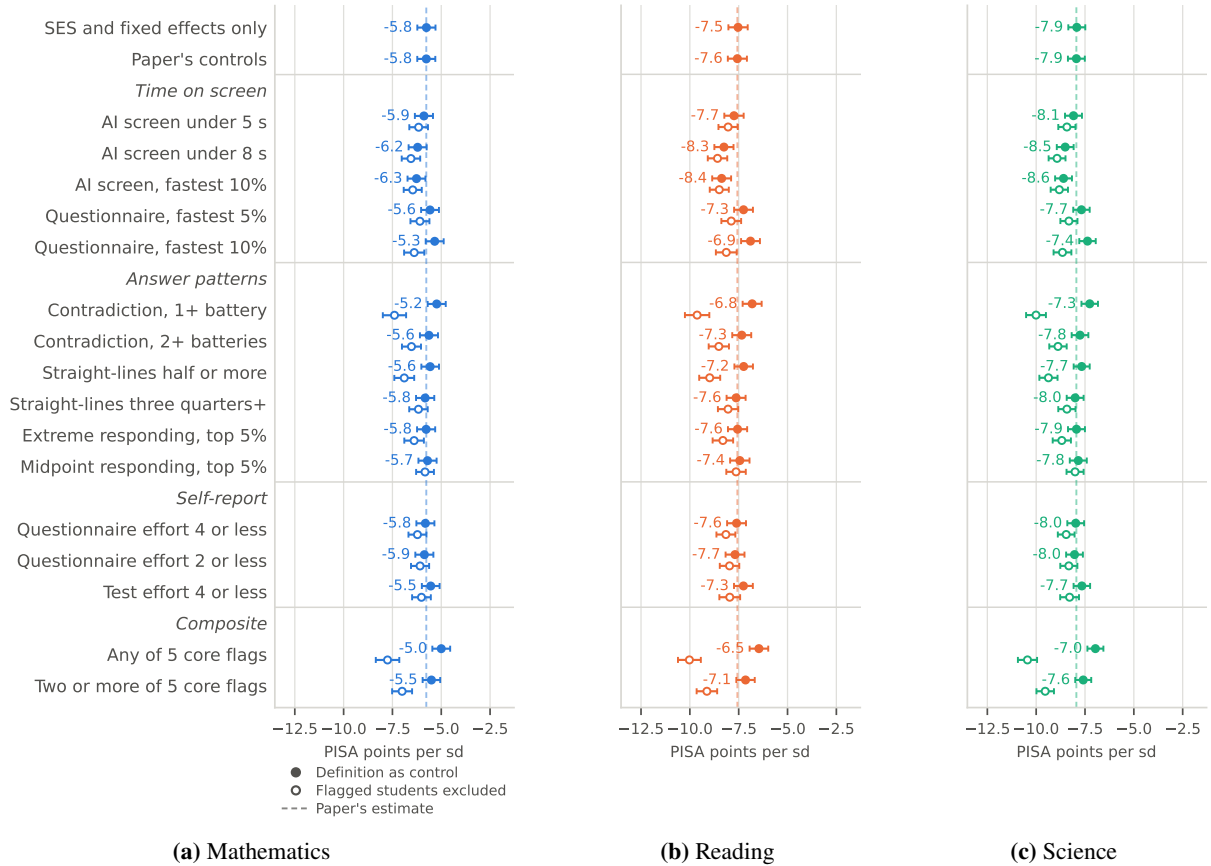
The third is the student's own report: at the end of the questionnaire students rate, from 1 to 10, the effort they put into answering it accurately, and the effort they put into the test. The fourth combines them: any, or two or more, of five core flags, namely under five seconds on the AI screen, the fastest twentieth on the questionnaire, a contradiction on any battery, straight-lining half the batteries, and questionnaire effort of 4 or less.

Table B.1 gives the share of students each definition flags, their mathematics gap relative to unflagged students in deviations from the country mean, and the share of the paper's AI-item straight-liners each one catches. Every definition flags low scorers: the gap is 41 to 75 points for all but extreme responding, where it is 21, and the definitions overlap little with one another.

Each definition is used two ways. It is added, as the only response-style term, to a specification with socioeconomic status and country or school fixed effects, and it is used to exclude the flagged students from that specification. For each we estimate the index alone, which is the level coefficient of Figure 2 of the paper, and the two principal components together, whose second coefficient is the purpose margin of Figure 4. Figures B.1 and B.2 plot the within-school estimates in the format of the paper's figures. Tables B.2 and B.3 report the exact values and add the country-fixed-effect estimates for mathematics; Table B.4 adds the paper's own indicator used on its own, the continuous versions of the measures, and the featured definitions combined with the paper's controls.

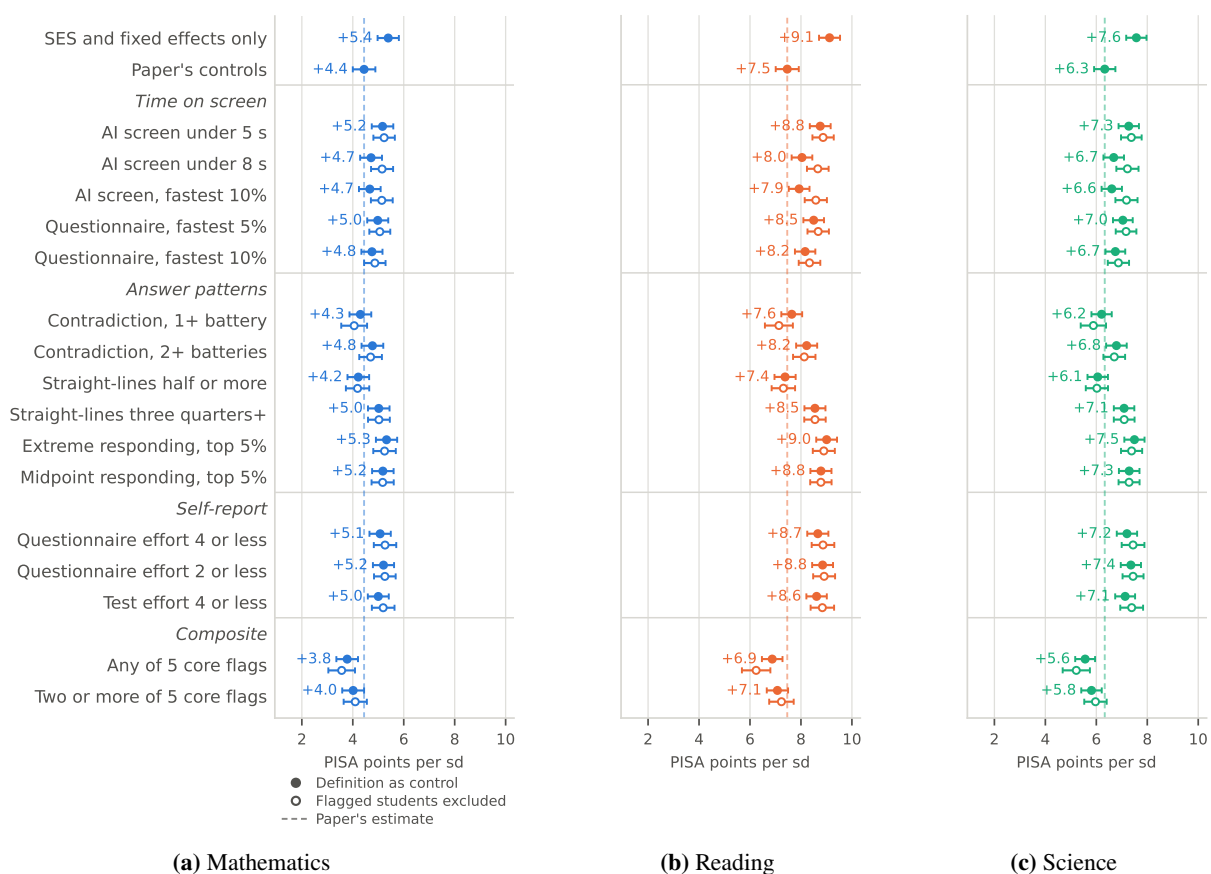
The estimates hold under every definition. Within schools, the level coefficient in mathematics lies between -4.9 and -7.8 across every external definition, against -5.8 in the paper, and the purpose coefficient between $+3.3$ and $+5.3$, against $+4.4$. Reading and science move in step. Excluding flagged students tends to steepen the level gradient rather than flatten it, because the students every definition flags score far below the rest and are spread across the index. The two loosest definitions, a contradiction on any battery and the composite of any flag, which drop 30 and 39 percent of the sample, are the ones that move the level coefficient most, and they move it further from zero.

Figure B.1: The level coefficient under alternative definitions of inattentive responding



Notes: Coefficient on the standardized chatbot index, in PISA points per standard deviation, with 95 percent intervals, from specifications with socioeconomic status and school fixed effects in which the row's definition of inattentive responding is either added as a control (filled markers) or used to exclude the flagged students (hollow markers), as the only response-style term. The dashed line is the paper's own within-school estimate, the last row of Figure 2. The first two rows are references: the specification with socioeconomic status and fixed effects only, and the paper's own specification. Definitions are in Table B.1; exact estimates, including country fixed effects, in Table B.2. Senate-weighted; standard errors combine 80 replicate weights with the imputation variance across ten plausible values.

Figure B.2: The purpose coefficient under alternative definitions of inattentive responding



Notes: Coefficient on the second principal component, learning against producing, from the two components entered together, in PISA points per standard deviation with 95 percent intervals. Specifications and markers as in Figure B.1; the dashed line is the paper's within-school estimate, the hollow PC2 marker of Figure 4. Exact estimates are in Table B.3.

Table B.1: Definitions of inattentive responding

Definition	Share flagged	Math gap	Of AI straight-liners
<i>Time on screen</i>			
Under 5 seconds on the AI screen	0.029	-70	0.04
Under 8 seconds on the AI screen	0.080	-63	0.10
Fastest 10% on the AI screen, within country	0.087	-59	0.12
Fastest 5% on the whole questionnaire, within country	0.050	-67	0.08
Fastest 10% on the whole questionnaire, within country	0.100	-53	0.16
<i>Answer patterns</i>			
Self-contradictory straight line on 1+ reverse-keyed battery	0.305	-45	0.48
Self-contradictory straight line on 2+ batteries	0.084	-56	0.18
Straight-lines half or more of 27 other batteries	0.118	-59	0.28
Straight-lines three quarters or more of them	0.023	-75	0.06
Extreme responding, top 5% within country	0.050	-21	0.07
Midpoint responding, top 5% within country	0.046	-41	0.09
<i>Self-report</i>			
Self-reported questionnaire effort 4 or less, of 10	0.088	-42	0.11
Self-reported questionnaire effort 2 or less, of 10	0.035	-51	0.05
Self-reported test effort 4 or less, of 10	0.117	-45	0.14
<i>Composite</i>			
Any of five core flags	0.389	-51	0.58
Two or more of five core flags	0.143	-64	0.29
<i>Built from the AI items, or a different sample</i>			
Same non-never answer to all four AI items	0.207	-24	1.00
Weekly AI use but ICT items say never or unavailable	0.059	-36	0.07

Notes: Senate-weighted shares on the estimation sample of 249,077 students. The mathematics gap is the mean score of flagged minus unflagged students, in deviations from the country mean. The last column is the share of the paper's straight-liners, students giving the same non-never answer to all four AI items, that the definition flags. Seconds are from the assessment's screen-timing file. The five batteries with both positively and negatively worded statements are ST034, ST131, ST324, ST307 and ST301; the 27 attitude and frequency batteries are ST034, ST092, ST094, ST097, ST100, ST129, ST131, ST216, ST265, ST300, ST301, ST307, ST322, ST326, ST432, ST433, ST435, ST436, ST468, ST469, ST495, ST496, ST498, ST499, ST500, ST501 and ST515. The cross-item contradiction is defined in the 24 countries that fielded the technology questionnaire.

Table B.2: The level coefficient under each definition: exact estimates

Definition	School fixed effects						Country FE	
	Math		Reading		Science		Math	
	Control	Excl.	Control	Excl.	Control	Excl.	Control	Excl.
<i>Reference</i>								
Socioeconomic status and fixed effects only	-5.76		-7.53		-7.92		-3.67	
	(0.24)		(0.25)		(0.22)		(0.31)	
Paper: straight line on the AI items, plus dispersion	-5.77		-7.57		-7.94		-3.78	
	(0.24)		(0.25)		(0.22)		(0.31)	
<i>Time on screen</i>								
Under 5 seconds on the AI screen	-5.89	-6.16	-7.74	-8.04	-8.09	-8.43	-3.90	-4.13
	(0.24)	(0.24)	(0.25)	(0.25)	(0.22)	(0.22)	(0.31)	(0.32)
Under 8 seconds on the AI screen	-6.21	-6.55	-8.25	-8.58	-8.51	-8.93	-4.40	-4.65
	(0.24)	(0.24)	(0.25)	(0.25)	(0.22)	(0.22)	(0.31)	(0.31)
Fastest 10% on the AI screen, within country	-6.27	-6.46	-8.37	-8.49	-8.60	-8.82	-4.49	-4.53
	(0.23)	(0.24)	(0.25)	(0.25)	(0.22)	(0.23)	(0.31)	(0.31)
Fastest 5% on the whole questionnaire, within country	-5.58	-6.09	-7.25	-7.88	-7.68	-8.32	-3.41	-4.05
	(0.23)	(0.25)	(0.25)	(0.26)	(0.21)	(0.22)	(0.30)	(0.31)
Fastest 10% on the whole questionnaire, within country	-5.34	-6.39	-6.90	-8.14	-7.37	-8.65	-3.08	-4.33
	(0.23)	(0.26)	(0.25)	(0.27)	(0.22)	(0.23)	(0.31)	(0.30)
<i>Answer patterns</i>								
Self-contradictory straight line on 1+ reverse-keyed battery	-5.23	-7.40	-6.81	-9.63	-7.26	-10.01	-3.03	-5.56
	(0.24)	(0.30)	(0.25)	(0.32)	(0.22)	(0.26)	(0.30)	(0.37)
Self-contradictory straight line on 2+ batteries	-5.63	-6.53	-7.34	-8.52	-7.76	-8.88	-3.54	-4.47
	(0.24)	(0.25)	(0.25)	(0.26)	(0.22)	(0.23)	(0.30)	(0.32)
Straight-lines half or more of 27 other batteries	-5.57	-6.89	-7.24	-8.98	-7.67	-9.37	-3.46	-4.89
	(0.23)	(0.26)	(0.24)	(0.27)	(0.21)	(0.24)	(0.29)	(0.32)
Straight-lines three quarters or more of them	-5.82	-6.17	-7.63	-8.04	-8.00	-8.43	-3.77	-4.11
	(0.24)	(0.24)	(0.25)	(0.27)	(0.22)	(0.23)	(0.31)	(0.32)
Extreme responding, top 5% within country	-5.77	-6.39	-7.56	-8.31	-7.94	-8.69	-3.70	-4.38
	(0.24)	(0.26)	(0.25)	(0.27)	(0.22)	(0.24)	(0.31)	(0.32)
Midpoint responding, top 5% within country	-5.70	-5.83	-7.44	-7.64	-7.84	-8.01	-3.59	-3.74
	(0.24)	(0.23)	(0.25)	(0.25)	(0.22)	(0.22)	(0.31)	(0.31)
<i>Self-report</i>								
Self-reported questionnaire effort 4 or less, of 10	-5.81	-6.22	-7.61	-8.16	-7.98	-8.46	-3.80	-4.29
	(0.24)	(0.23)	(0.25)	(0.25)	(0.22)	(0.22)	(0.30)	(0.32)
Self-reported questionnaire effort 2 or less, of 10	-5.87	-6.09	-7.69	-7.97	-8.04	-8.33	-3.86	-4.06
	(0.24)	(0.24)	(0.25)	(0.25)	(0.22)	(0.22)	(0.31)	(0.31)
Self-reported test effort 4 or less, of 10	-5.54	-6.01	-7.26	-7.96	-7.66	-8.29	-3.46	-4.04
	(0.23)	(0.24)	(0.25)	(0.27)	(0.22)	(0.25)	(0.30)	(0.32)
<i>Composite</i>								
Any of five core flags	-5.00	-7.75	-6.46	-10.02	-6.97	-10.44	-2.77	-5.96
	(0.23)	(0.31)	(0.24)	(0.30)	(0.21)	(0.25)	(0.29)	(0.38)
Two or more of five core flags	-5.50	-7.00	-7.15	-9.13	-7.59	-9.54	-3.40	-5.15
	(0.23)	(0.26)	(0.24)	(0.27)	(0.21)	(0.23)	(0.29)	(0.32)

Notes: Coefficient on the standardized chatbot index entered alone, in PISA points per standard deviation, standard errors in parentheses. Each row's definition enters as a control (Control) or excludes the flagged students (Excl.) in a specification with socioeconomic status and the fixed effects shown, with the definition as the only response-style term. The first three domain pairs use school fixed effects, as plotted in Figure B.1; the last pair uses country fixed effects.

Table B.3: The purpose coefficient under each definition: exact estimates

Definition	School fixed effects						Country FE	
	Math		Reading		Science		Math	
	Control	Excl.	Control	Excl.	Control	Excl.	Control	Excl.
<i>Reference</i>								
Socioeconomic status and fixed effects only	5.39		9.12		7.57		8.98	
	(0.21)		(0.21)		(0.20)		(0.25)	
Paper: straight line on the AI items, plus dispersion	4.44		7.46		6.33		7.15	
	(0.23)		(0.23)		(0.21)		(0.26)	
<i>Time on screen</i>								
Under 5 seconds on the AI screen	5.17	5.23	8.76	8.87	7.27	7.37	8.47	8.65
	(0.22)	(0.21)	(0.21)	(0.21)	(0.20)	(0.21)	(0.24)	(0.25)
Under 8 seconds on the AI screen	4.72	5.15	8.04	8.66	6.68	7.22	7.67	8.42
	(0.22)	(0.22)	(0.21)	(0.22)	(0.21)	(0.22)	(0.24)	(0.25)
Fastest 10% on the AI screen, within country	4.67	5.14	7.93	8.58	6.61	7.18	7.59	8.39
	(0.22)	(0.22)	(0.21)	(0.22)	(0.20)	(0.22)	(0.24)	(0.25)
Fastest 5% on the whole questionnaire, within country	4.98	5.06	8.50	8.68	7.04	7.17	8.18	8.37
	(0.21)	(0.21)	(0.21)	(0.21)	(0.20)	(0.21)	(0.24)	(0.24)
Fastest 10% on the whole questionnaire, within country	4.75	4.87	8.16	8.34	6.75	6.87	7.83	8.09
	(0.21)	(0.21)	(0.20)	(0.22)	(0.20)	(0.21)	(0.24)	(0.24)
<i>Answer patterns</i>								
Self-contradictory straight line on 1+ reverse-keyed battery	4.30	4.05	7.64	7.13	6.21	5.88	7.12	6.91
	(0.22)	(0.26)	(0.21)	(0.28)	(0.20)	(0.25)	(0.25)	(0.29)
Self-contradictory straight line on 2+ batteries	4.77	4.70	8.22	8.13	6.79	6.70	7.87	7.92
	(0.22)	(0.23)	(0.21)	(0.22)	(0.21)	(0.22)	(0.25)	(0.25)
Straight-lines half or more of 27 other batteries	4.22	4.19	7.38	7.31	6.06	6.02	7.03	7.09
	(0.22)	(0.23)	(0.21)	(0.23)	(0.20)	(0.22)	(0.25)	(0.26)
Straight-lines three quarters or more of them	5.02	5.03	8.55	8.55	7.09	7.09	8.33	8.36
	(0.22)	(0.22)	(0.21)	(0.21)	(0.21)	(0.21)	(0.24)	(0.25)
Extreme responding, top 5% within country	5.32	5.25	9.01	8.90	7.50	7.38	8.85	8.76
	(0.21)	(0.23)	(0.21)	(0.22)	(0.20)	(0.21)	(0.25)	(0.26)
Midpoint responding, top 5% within country	5.18	5.17	8.78	8.78	7.29	7.29	8.62	8.65
	(0.22)	(0.22)	(0.21)	(0.21)	(0.21)	(0.21)	(0.25)	(0.25)
<i>Self-report</i>								
Self-reported questionnaire effort 4 or less, of 10	5.07	5.27	8.66	8.86	7.20	7.44	8.42	8.71
	(0.21)	(0.22)	(0.21)	(0.23)	(0.20)	(0.23)	(0.24)	(0.26)
Self-reported questionnaire effort 2 or less, of 10	5.20	5.26	8.85	8.91	7.36	7.44	8.64	8.77
	(0.21)	(0.22)	(0.21)	(0.22)	(0.20)	(0.21)	(0.24)	(0.25)
Self-reported test effort 4 or less, of 10	5.00	5.19	8.61	8.84	7.13	7.39	8.37	8.73
	(0.21)	(0.23)	(0.20)	(0.24)	(0.20)	(0.23)	(0.24)	(0.26)
<i>Composite</i>								
Any of five core flags	3.78	3.56	6.87	6.24	5.56	5.21	6.26	5.98
	(0.22)	(0.27)	(0.21)	(0.28)	(0.20)	(0.27)	(0.25)	(0.30)
Two or more of five core flags	4.01	4.10	7.08	7.23	5.81	5.97	6.62	6.88
	(0.22)	(0.23)	(0.21)	(0.25)	(0.21)	(0.22)	(0.25)	(0.26)

Notes: Coefficient on the second principal component from the two entered together, in PISA points per standard deviation, standard errors in parentheses. Specifications as in Table B.2.

Table B.4: Further specifications: the paper's own indicator, continuous measures, and combinations

Specification	Level coefficient				Purpose coefficient			
	School FE			Country FE	School FE			Country FE
	Math	Reading	Science	Math	Math	Reading	Science	Math
<i>Built from the AI items</i>								
Same non-never answer to all four AI items, as control	-2.83	-2.88	-4.00	0.95	4.28	7.39	6.11	7.16
	(0.26)	(0.29)	(0.25)	(0.32)	(0.22)	(0.22)	(0.21)	(0.26)
Same non-never answer to all four AI items, excluded	-4.46	-5.18	-6.24	-0.83	4.31	7.42	6.17	7.27
	(0.30)	(0.36)	(0.29)	(0.34)	(0.23)	(0.23)	(0.22)	(0.26)
<i>ICT-questionnaire countries</i>								
Socioeconomic status and fixed effects only	-7.25	-8.74	-9.54	-4.95	4.85	7.57	6.49	8.53
	(0.33)	(0.31)	(0.30)	(0.45)	(0.31)	(0.32)	(0.33)	(0.33)
Weekly AI use but ICT items say never or un-available, as control	-6.65	-8.18	-8.91	-3.98	5.07	7.79	6.74	8.88
	(0.33)	(0.31)	(0.30)	(0.46)	(0.31)	(0.32)	(0.33)	(0.33)
Same, excluded	-6.50	-7.92	-8.70	-3.66	5.27	8.10	7.03	9.31
	(0.34)	(0.33)	(0.31)	(0.48)	(0.32)	(0.34)	(0.36)	(0.33)
<i>Continuous measures, as controls</i>								
Straight-lining share, continuous	-5.28	-6.81	-7.29	-3.10	3.35	6.10	4.91	5.71
	(0.23)	(0.24)	(0.21)	(0.30)	(0.22)	(0.21)	(0.21)	(0.25)
Self-contradictory straight lines, count	-5.40	-7.02	-7.46	-3.27	4.20	7.45	6.08	6.94
	(0.24)	(0.25)	(0.22)	(0.30)	(0.22)	(0.21)	(0.20)	(0.25)
Log seconds on the AI screen	-6.63	-8.91	-9.06	-5.05	3.83	6.69	5.55	6.32
	(0.24)	(0.25)	(0.22)	(0.31)	(0.22)	(0.21)	(0.21)	(0.25)
Log seconds per item, whole questionnaire	-4.94	-6.34	-6.83	-2.58	4.74	8.17	6.70	7.96
	(0.24)	(0.25)	(0.22)	(0.30)	(0.21)	(0.21)	(0.20)	(0.25)
Questionnaire effort, 1 to 10	-5.71	-7.45	-7.85	-3.68	4.79	8.19	6.84	8.04
	(0.24)	(0.25)	(0.22)	(0.31)	(0.21)	(0.21)	(0.20)	(0.25)
<i>Combined with the paper's controls</i>								
Paper's controls plus fastest 10% on the AI screen, within country	-6.26	-8.38	-8.60	-4.54	3.96	6.66	5.68	6.22
	(0.23)	(0.24)	(0.22)	(0.30)	(0.23)	(0.23)	(0.21)	(0.26)
Paper's controls plus self-contradictory straight line on 2+ batteries	-5.64	-7.38	-7.78	-3.63	4.33	7.30	6.19	6.88
	(0.23)	(0.25)	(0.22)	(0.30)	(0.23)	(0.23)	(0.22)	(0.26)
Paper's controls plus two or more of five core flags	-5.51	-7.19	-7.61	-3.46	3.95	6.73	5.70	6.20
	(0.23)	(0.24)	(0.21)	(0.29)	(0.23)	(0.23)	(0.21)	(0.26)
Paper's controls, fastest 10% on the AI screen, within country excluded	-6.36	-8.35	-8.69	-4.46	4.01	6.70	5.75	6.39
	(0.23)	(0.25)	(0.22)	(0.30)	(0.24)	(0.24)	(0.23)	(0.27)
Paper's controls, self-contradictory straight line on 2+ batteries excluded	-6.52	-8.54	-8.89	-4.54	4.29	7.26	6.14	6.97
	(0.25)	(0.26)	(0.23)	(0.31)	(0.24)	(0.24)	(0.22)	(0.27)
Paper's controls, two or more of five core flags excluded	-7.01	-9.16	-9.55	-5.22	3.99	6.78	5.79	6.36
	(0.26)	(0.27)	(0.23)	(0.32)	(0.25)	(0.26)	(0.23)	(0.29)

Notes: PISA points per standard deviation, standard errors in parentheses; level and purpose coefficients as in Tables B.2 and B.3. The first block uses the paper's own straight-lining indicator on its own. The ICT block is estimated on the 129,206 students in the 24 countries that fielded the technology questionnaire, and its first row is that sample's estimate with socioeconomic status and fixed effects only. The continuous measures enter as controls, with a missing indicator where needed. The last block adds each featured definition to the paper's full specification, or excludes the students it flags from it.

C Alternative control sets

The paper's specification conditions on socioeconomic status, the response-style terms and country or school fixed effects. This appendix adds two further blocks of controls, cumulatively. The first is pre-determined: gender, grade relative to the modal grade, grade repetition, immigrant background, language at home, age, and the assessment's indices of ICT resources at home and outside school. The second measures engagement and effort: effort on the test, perseverance, curiosity, self-regulation, goal setting, enjoyment of problem solving, cognitive adaptability, expected education, homework time, arriving late and skipping school. The second block sits inside the mechanism of Section 2 of the paper, where AI substitutes for the student's effort, so estimates that condition on it bound the substitution channel from below rather than sharpen it; that is why it is a robustness row rather than the headline specification. Missing values on a control are mean-filled with an indicator, so every row uses the paper's 249,077 students.

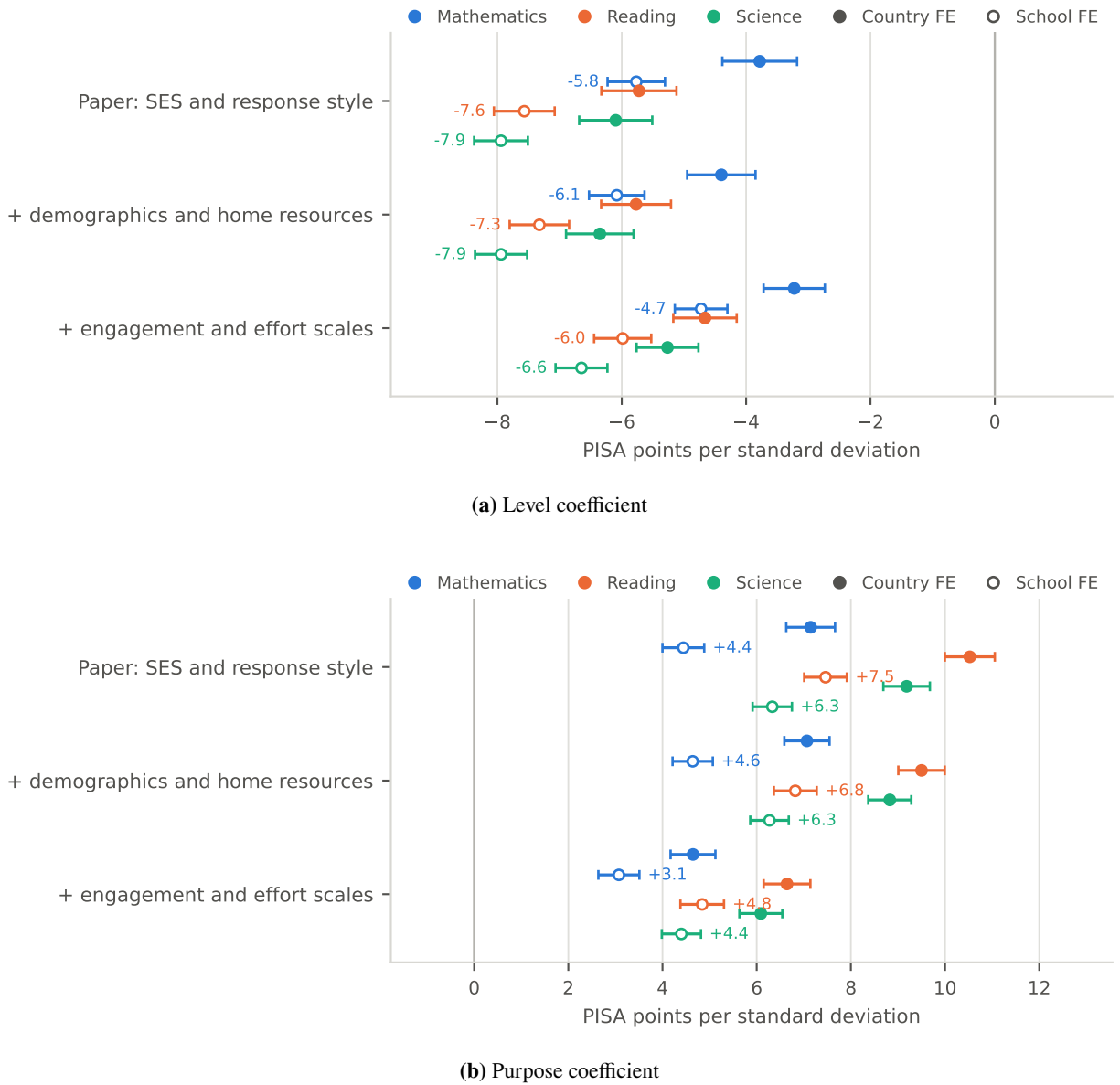
Figure C.1 shows the level and purpose coefficients for the three control sets under country and school fixed effects, and Table C.1 reports them for mathematics together with the four items. The demographic block leaves every estimate within a third of a point: within schools the level coefficient moves from -5.77 to -6.08 in mathematics and from -7.57 to -7.32 in reading, and the purpose coefficient from $+4.44$ to $+4.64$. The engagement block lowers the level coefficient to -4.72 within schools and the purpose coefficient to $+3.07$, and moves the drafting item from -5.51 to -3.54 and the learning item from $+2.18$ to $+1.35$; every sign holds and every interval excludes zero.

Table C.1: Alternative control sets: exact estimates, mathematics

	Level	Summarize	Draft	Research	Learn	Purpose (PC2)
<i>Panel A. Country fixed effects</i>						
Paper: SES and response style	-3.78 (0.31)	-5.93 (0.35)	-7.58 (0.38)	3.21 (0.38)	6.32 (0.40)	7.15 (0.26)
+ demographics and home resources	-4.40 (0.28)	-6.08 (0.33)	-7.46 (0.36)	2.66 (0.36)	6.20 (0.36)	7.07 (0.24)
+ engagement and effort scales	-3.23 (0.25)	-4.30 (0.32)	-4.71 (0.36)	1.51 (0.33)	4.05 (0.35)	4.65 (0.24)
<i>Panel B. School fixed effects</i>						
Paper: SES and response style	-5.77 (0.24)	-5.99 (0.31)	-5.51 (0.33)	2.85 (0.30)	2.18 (0.32)	4.44 (0.23)
+ demographics and home resources	-6.08 (0.23)	-6.18 (0.31)	-5.54 (0.32)	2.44 (0.29)	2.48 (0.31)	4.64 (0.22)
+ engagement and effort scales	-4.72 (0.22)	-4.76 (0.30)	-3.54 (0.31)	1.62 (0.29)	1.35 (0.31)	3.07 (0.22)

Notes: Mathematics, PISA points per standard deviation, standard errors in parentheses; 249,077 students in 35 countries in every row. Level is the index entered alone; the four items are entered together, each holding the other three fixed; Purpose is the second component from the two entered together. Rows add the demographic and engagement blocks cumulatively to the paper's specification.

Figure C.1: The level and purpose coefficients under alternative control sets



Notes: PISA points per standard deviation, with 95 percent intervals that combine 80 replicate weights with the imputation variance across ten plausible values. Rows add the two control blocks cumulatively to the paper's specification; within a row, filled markers use country fixed effects and hollow markers school fixed effects, and the labels give the within-school estimates. Panel (a) is the coefficient on the chatbot index entered alone, the level margin of Figure 2; panel (b) the coefficient on the second principal component from the two entered together, the purpose margin of Figure 4. Exact estimates for mathematics, including the four items, are in Table C.1.

D The AI items

This appendix reproduces the wording of the eight questions on artificial intelligence in the PISA 2025 student questionnaire, with their answer scales, taken from the variable labels of the public-use file. Table D.1 gives the weighted distribution of answers.

Chatbot use: the four items of the headline index

Stem: “How often do you use Artificial Intelligence chatbots (e.g. <ChatGPT>) for your school work to do the following activities?”

- ST438Q01DA (1): “To summarize a text I had to read”
- ST438Q02DA (2): “To conduct preliminary research on a new topic”
- ST438Q03DA (3): “To draft texts for writing assignments”
- ST438Q04DA (4): “To help me learn”

Response options. 1 = Never or almost never; 2 = About once or twice a year; 3 = About once or twice a month; 4 = About once or twice a week; 5 = Every day or almost every day.

AI tool use: the two items added in the broad index

Stem: “This school year, how often did you use the following <digital resources>at school?”

- IC170Q10DA (5): “<Artificial Intelligence (AI)>tools or software (e.g. <ChatGPT>, <MidJourney>, <Stable Diffusion>)”

Stem: “This school year, how often did you use the following <digital resources>outside of school (e.g., at home or where you usually access <digital resources>)?”

- IC171Q10DA (6): “<Artificial Intelligence (AI)>tools or software (e.g. <ChatGPT>, <MidJourney>, <Stable Diffusion>)”

Response options for (5). 1 = Never or almost never; 2 = About once or twice a month; 3 = About once or twice a week; 4 = Every day or almost every day; 5 = Several times a day; 6 = This resource is not available to me at school.

Response options for (6). 1 = Never or almost never; 2 = About once or twice a month; 3 = About once or twice a week; 4 = Every day or almost every day; 5 = Several times a day; 6 = This resource is not available to me outside school.

Items considered and excluded from the index

Stem: “This school year, how often did you do the following tasks using <digital resources>in your school lessons?”

- ST436Q16DA (7): “Assess the quality of information generated by Artificial Intelligence chatbots (e.g. <ChatGPT>, <Google Bard>)”

Stem: “To what extent do you agree or disagree with the following statements?”

- IC182Q05DA (8): “I am interested in learning about <machine learning>and <artificial intelligence>.”

Response options for (7). 1 = Never or almost never; 2 = Sometimes; 3 = Often; 4 = Very often. *Response options for (8).* 1 = Strongly disagree; 2 = Disagree; 3 = Agree; 4 = Strongly agree.

Notes: Item (6) is stored in the public-use file with its text cut off mid-phrase; it repeats item (5) verbatim and differs only in the stem, which asks about use outside school. Angle brackets mark placeholders that each country replaces with a locally appropriate term.

Table D.1: Weighted response distribution of the eight AI items

Item	Scale	Weighted % giving each response					Countries	N
		1	2	3	4	5		
Summarize a text (1)	1–5	32.5	16.5	21.3	19.0	10.7	38	275,812
Preliminary research (2)	1–5	28.8	16.8	23.1	20.2	11.1	38	274,747
Draft writing (3)	1–5	32.2	16.7	22.1	18.1	10.8	38	274,220
Help me learn (4)	1–5	20.7	13.1	20.6	26.4	19.1	38	273,934
AI tools at school (5)	1–5	32.0	15.7	19.7	16.7	15.9	24	144,140
AI tools outside school (6)	1–5	28.7	13.3	21.1	18.8	18.1	24	143,456
Assess AI output (7)	1–4	37.3	28.5	20.6	13.5	–	38	256,512
Interest in ML/AI (8)	1–4	19.0	31.0	38.5	11.4	–	24	148,182

Notes: OECD sample, senate-weighted. Items (1) to (4) run never / once or twice a year / once or twice a month / once or twice a week / every day. Items (5) and (6) run never / monthly / weekly / daily / several times a day, and their sixth response, “this resource is not available to me”, is treated as missing rather than as a frequency. Items (7) and (8) use four-point scales.

E Index construction

This appendix shows how the chatbot index and the purpose component are built from the answers. Table E.1 gives the correlations among the AI items on which the principal components rest.

Let x_{ij} be student i ’s answer to chatbot question j , on the 1 to 5 scale, and w_i the student’s senate weight. Each answer is standardized with its weighted mean $\bar{x}_j$ and standard deviation s_j across OECD students, $z_{ij} = (x_{ij} - \bar{x}_j)/s_j$, and the four standardized answers form the vector $\mathbf{z}_i$. Their weighted correlation matrix and its eigenvectors are

$$R = \frac{\sum_i w_i \mathbf{z}_i \mathbf{z}_i'}{\sum_i w_i}, \quad R \mathbf{v}_k = \lambda_k \mathbf{v}_k, \quad \lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4,$$

and student i ’s score on component k is

$$f_{ik} = \frac{\mathbf{z}_i' \mathbf{v}_k}{\sqrt{\lambda_k}},$$

which has weighted mean zero and standard deviation one. The chatbot index is f_{i1} and the purpose component f_{i2} , with the sign of each eigenvector chosen so that higher values mean more AI use and use leaning more toward learning. The loading of question j on component k , its correlation with the score, is $v_{jk} \sqrt{\lambda_k}$.

Table E.1: Weighted correlations among the AI items

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Summarize a text (1)	1							
Preliminary research (2)	0.68	1						
Draft writing (3)	0.69	0.67	1					
Help me learn (4)	0.60	0.65	0.61	1				
AI tools at school (5)	0.40	0.40	0.40	0.42	1			
AI tools outside school (6)	0.40	0.42	0.40	0.46	0.61	1		
Assess AI output (7)	0.31	0.34	0.31	0.35	0.41	0.38	1	
Interest in ML/AI (8)	0.12	0.15	0.11	0.16	0.13	0.21	0.16	1

Notes: OECD sample, senate-weighted, pairwise complete. Items (1) to (4) are the chatbot battery, (5) and (6) the technology questionnaire, (7) and (8) the two items excluded from the index.

We decompose the correlation rather than the covariance matrix because the two item families run on different frequency scales, and an unstandardized decomposition would let the wider-variance items dominate for no substantive reason (Jolliffe, 2002). Pearson correlations on five-point items are standard practice here;

a polychoric matrix would raise the loadings somewhat without materially reordering students. Estimation is listwise on the set's items.

In terms of the raw answers, the two components are

$$\begin{aligned} \text{PC1}_i &= -2.29 + 0.21 S_i + 0.22 P_i + 0.21 D_i + 0.20 L_i, \\ \text{PC2}_i &= -0.52 - 0.47 S_i + 0.06 P_i - 0.45 D_i + 0.88 L_i, \end{aligned}$$

where S_i , P_i , D_i and L_i are the answers, from 1 to 5, to the questions on summarizing a text, preliminary research, drafting writing and help with learning. The first gives the four questions almost equal weights, which is why it behaves like their sum. The second puts help with learning against summarizing and drafting, with research close to zero, so it measures how far a student's use leans toward learning rather than producing work at a given level of use. Its eigenvalue, 0.42, is below the rule that retains only the first component (Kaiser, 1960); we use it as the contrast that Section 2 of the paper specifies in advance rather than as a second factor, and Section 6 of the paper reaches the same conclusion from the four questions entered together. Estimated separately in each of the 38 countries, the first component explains 67 to 81 percent of the variance, and the second puts learning against summarizing and drafting in 33 of them; in the other five, Finland, Israel, Japan, Poland and Slovakia, drafting loads close to zero.

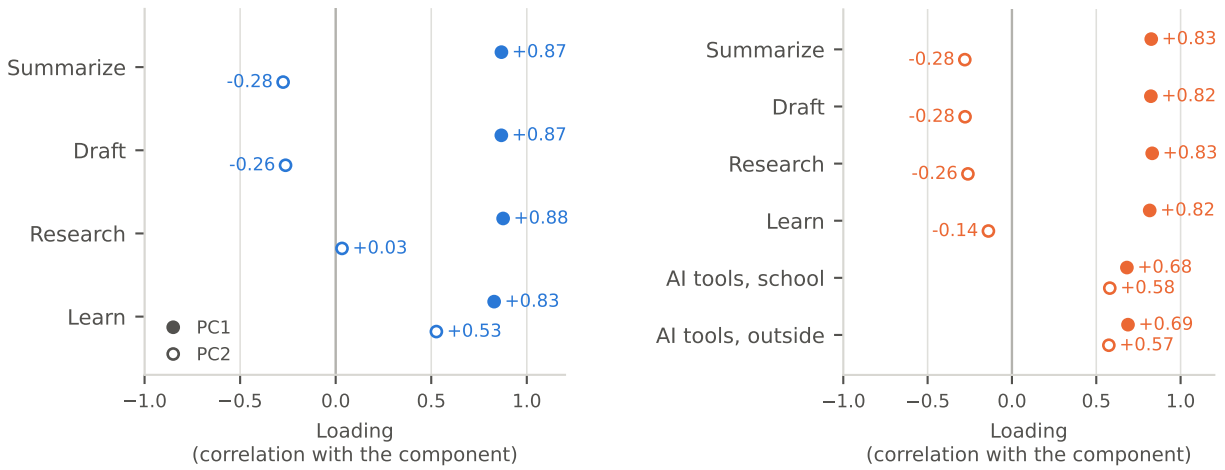
Figure E.2 places the index next to the one the OECD distributes with the assessment, a weighted-likelihood score built from the same four items. Both are functions of the response pattern alone, so the 625 patterns map one index onto the other with a correlation of 0.979, the OECD score stretching the two extremes; and the adjusted mathematics gradient is the same on either scale. Replacing our index with the OECD's in the paper's specifications gives -4.02 , -6.04 and -6.40 points per standard deviation within countries and -5.81 , -7.66 and -8.02 within schools, against -3.78 , -5.72 , -6.10 and -5.77 , -7.57 , -7.94 with ours.

Table E.2: Principal component loadings, both item sets

Item	Chatbot set		Broad set	
	PC1	PC2	PC1	PC2
Summarize a text	0.867	-0.275	0.826	-0.280
Preliminary research	0.876	0.033	0.832	-0.261
Draft writing	0.867	-0.262	0.824	-0.278
Help me learn	0.829	0.527	0.817	-0.139
AI tools at school	–	–	0.682	0.580
AI tools outside school	–	–	0.688	0.575
Eigenvalue	2.958	0.423	3.658	0.910
Share of variance	74.0%	10.6%	61.0%	15.2%
Cronbach's α	0.882		0.870	
Students	271,298		128,323	
Countries	38		24	

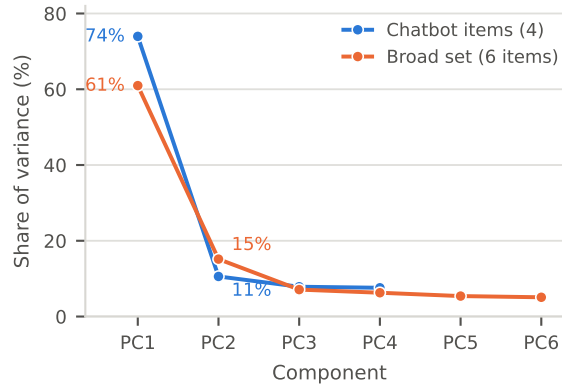
Notes: Weighted PCA on the correlation matrix, senate weights. The broad set is estimated on the countries fielding the technology questionnaire.

Figure E.1: Loadings and variance explained, weighted principal components



(a) Chatbot items, 38 countries

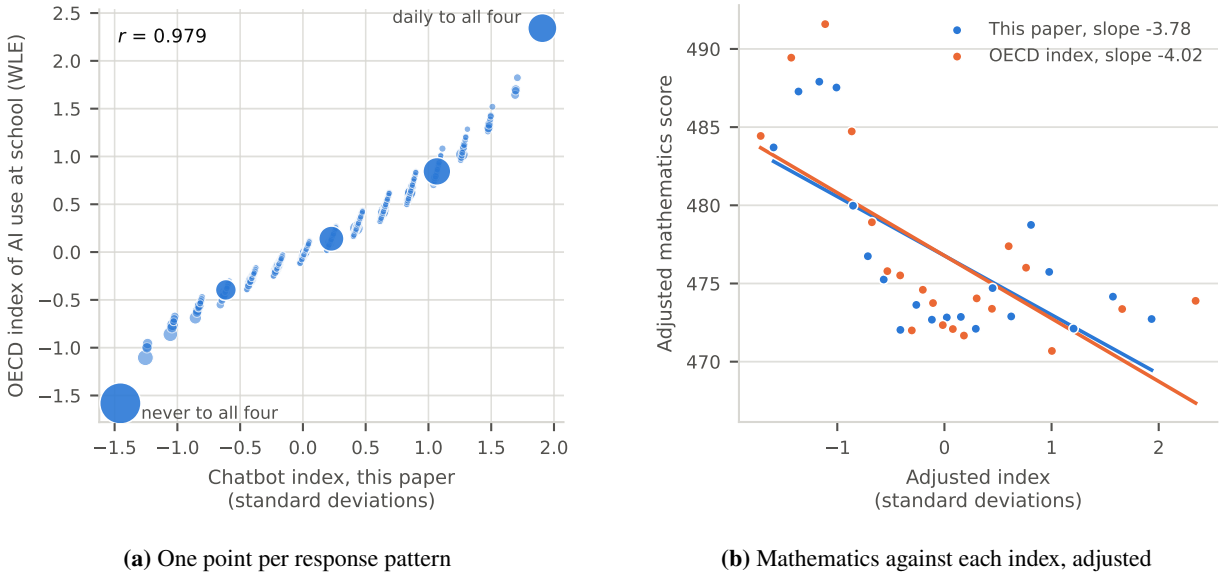
(b) Broad set, 24 countries



(c) Variance explained, both sets

Notes: OECD countries, senate-weighted principal components of the item correlation matrix. Panels (a) and (b) give each item's loading, its correlation with the component, on the first (filled) and second (hollow) components; the broad set adds the two AI-tool items from the optional questionnaire. Panel (c) gives the share of variance each component explains. The chatbot items load almost equally on the first component, which is what makes it close to the item sum and therefore exposed to the response-style problem of Section 3 of the paper.

Figure E.2: The chatbot index against the index distributed with the assessment



Notes: Panel (a): both indices are functions of the four chatbot answers alone, so every student with a given response pattern has the same value on each; the panel draws one point per pattern, 625 in all, with area proportional to the senate-weighted share of students giving it, the paper's first principal component on the horizontal axis and the OECD's weighted-likelihood index of AI use at school on the vertical. Panel (b): residualized binned scatter of mathematics on each index, both standardized to unit variance and both adjusted, with the score, for country fixed effects, socioeconomic status and the response-style terms, as in Figure 3 of the paper; twenty equally weighted bins per index, lines at the fitted slopes. OECD countries, 249,077 students in the regression sample of the paper.

F The response-style artifact

This appendix documents the response-style problem described in Section 3 of the paper. Table F.1 compares students who give the same answer to all four chatbot questions with students who reach the same total through different answers, both measured against their country's mean mathematics score.

Table F.1: Straight-liners and mixed responders at identical reported AI use

Item total	Straight-lined	Mixed	Difference	<i>N</i> straight	<i>N</i> mixed
8	-37.4	+21.8	59.3	8,650	12,451
12	-16.4	+14.9	31.4	13,076	13,637
16	-1.1	+17.2	18.3	16,598	7,034

Notes: Mathematics score in deviations from the student's own country mean, senate-weighted. The item total is the sum of the four chatbot items, so 8 corresponds for instance to answering 2 on all four or to any mixed combination summing to 8. Totals of 4 and 20 admit only straight-lined answers and so have no comparison group; totals not divisible by four admit none.

The dips at the straight-line totals are a within-country phenomenon: the standard deviation of the country-composition index across bins is an order of magnitude smaller than that of the bin means themselves, and it survives demeaning by country.

It is also more general than straight-lining as such. A binned scatter drawn against the index stays jagged after straight-liners are dropped, and adjusting further for within-student response dispersion reduces the jaggedness by about a third. The deeper reason is that the index has lumpy discrete support, four items with five categories each, so which response patterns fall into a given bin varies from bin to bin, and patterns carry achievement information of their own: among non-straight-liners, bin-level mean response dispersion

correlates about -0.5 with bin-level mean mathematics. Residualizing both axes on continuous controls, as in Section 4 of the paper, is what solves it.

G Sample

This appendix lists the countries in the sample. The paper uses all 38 OECD members that took part in PISA 2025; Table G.1 gives the number of students assessed in each, and its note explains how the estimation samples are reached.

Table G.1: The 38 OECD countries in PISA 2025, with students assessed

Country	<i>N</i>	Country	<i>N</i>	Country	<i>N</i>	Country	<i>N</i>
AUS	14,658	DNK	5,098	ISR	7,330	NZL	5,004
AUT	6,987	ESP	29,966	ITA	8,946	POL	7,103
BEL	7,456	EST	6,304	JPN	6,554	PRT	6,945
CAN	24,601	FIN	7,668	KOR	6,859	SVK	6,682
CHE	7,597	FRA	6,501	LTU	7,718	SVN	7,366
CHL	6,573	GBR	15,546	LUX	6,078	SWE	6,688
COL	7,268	GRC	6,858	LVA	6,968	TUR	7,702
CRI	6,881	HUN	6,387	MEX	7,843	USA	4,623
CZE	8,462	IRL	6,356	NLD	6,636		
DEU	6,659	ISL	3,974	NOR	6,469		
Total							315,314

Notes: Countries are identified by ISO code and selected on the file’s own OECD indicator. *N* is the number of assessed students in the public-use student file. Estimation samples are smaller because they require valid responses to the relevant items: the chatbot index is defined for 271,298 students, and the main regression sample, which additionally requires socioeconomic status and a school identifier, is 249,077 in 35 countries. Costa Rica drops for want of the socioeconomic index, and Iceland and Luxembourg for want of school identifiers, both being systems that assess close to their entire eligible population. The two technology items were fielded in 24 of the 38.

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